

🕶️ Language-Queryable 3D Scene Understanding

Depth-Anything V2 · CLIP ViT-B/32 · FastSAM · Llama 4 Vision

Overall Score	Points	Mask Regions	Pose Quality
58.6/100	90,775	302	99/100

🕶️ 3D Scene 🔍 Query Scene 📄 Scene Info 📊 Accuracy

Accuracy Report

Overall Score

58.6/100

Depth Consistency	Pose Quality	Query Quality	CLIP Coverage
0.0/100	99.2/100	73.7/100	53.2/100
↑ median err: 78.9cm	↑ PnP: 98.7%	↑ conf: 51.7%	↑ 44.9% covered

Per-Query Breakdown

	query	confidence	compactness	precision
0	floor	61.1	0.865	
1	wall	49	0.711	
2	navigable path	41.7	0.912	
3	obstacle	48.6	0.789	
4	furniture	58	0.707	

Full Report (JSON)

```
{
  "overall_score" : 58.6
  "depth_consistency" : {
    "mean_error_m" : 1.067
    "median_error_m" : 0.7888
    "pct_under_10cm" : 6
    "pct_under_20cm" : 11.9
    "score" : 0
    "n_measurements" : 1290
  }
  "pose_quality" : {
    "pnp_success_pct" : 98.7
  }
}
```

```
"smoothness_score" : 100
"trajectory_length" : 7.352
"mean_step_m" : 0.0955
"max_step_m" : 0.4081
"n_trajectory_jumps" : 0
"score" : 99.2
}
"query_quality" : {
  "mean_confidence" : 51.7
  "mean_compactness" : 0.797
  "mean_precision_at_k" : 0.972
  "score" : 73.7
  "per_query" : [
    0 : {
      "query" : "floor"
      "confidence" : 61.1
      "compactness" : 0.865
      "precision_at_k" : 0.922
      "top_k" : 500
    }
    1 : {
      "query" : "wall"
      "confidence" : 49
      "compactness" : 0.711
      "precision_at_k" : 1
      "top_k" : 500
    }
    2 : {
      "query" : "navigable path"
      "confidence" : 41.7
      "compactness" : 0.912
      "precision_at_k" : 0.94
      "top_k" : 500
    }
    3 : {
      "query" : "obstacle"
      "confidence" : 48.6
      "compactness" : 0.789
      "precision_at_k" : 1
      "top_k" : 500
    }
    4 : {
      "query" : "furniture"
      "confidence" : 58
      "compactness" : 0.707
      "precision_at_k" : 1
      "top_k" : 500
    }
  ]
}
"embedding_coverage" : {
  "total_points" : 90775
  "assigned_points" : 40775
}
```

```
}  
  "coverage_pct" : 44.9  
  "n_mask_regions" : 302  
  "score" : 53.2  
}
```